Operator-Entropy Joint-Embedding Predictive Architectures for Multivariate Time-Series Anomaly Detection: Latent Residual Conditioning and Calibrated Thresholding

Andre Williams, Imadeldin Mahgoub, *Senior Member, IEEE,* and Hamid Akbarian

***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize observation-space error, fitting stochastic sensor noise that produces heavy-tailed residuals and severe false-alarm flooding under automated threshold calibration. We investigate Joint-Embedding Predictive Architectures (JEPAs) that score discrepancies in a low-dimensional latent space, together with an Operator-Entropy variant (OpEntropy JEPA) whose linear Koopman transition operator is constrained by Von Neumann spectral entropy and a shifted logarithmic Bregman energy barrier, paired with coordinate saliency gating and a Fourier-domain dyadic frequency front-end. Across 33 continuous telemetry streams from seven physical datasets under chronological held-out Streaming Peaks-Over-Threshold calibration, latent predictive scoring suppresses false alarms four-fold under retrospective buffered attribution (7.6% to 8.6% nominal false-positive rate versus 30.0% for TimesNet and 30.7% for TranAD) and twelve- to eighteen-fold under strictly causal streaming (1.4% to 1.9% versus 22.2% to 24.3%). This operational false-alarm reduction does not establish general detection superiority: point-level F1, precision-recall ranking, and matched realized-operating-point event recall remain statistically comparable to reconstructive baselines. Controlled mechanism ablations isolate two distinct drivers: low-dimensional LayerNorm bottlenecks mechanically compress residual kurtosis, while learned latent projections halve inter-block distribution drift relative to normalized observation forecasters. Factorial sweeps further show that spectral entropy and barrier terms preserve full operator rank and near-unit condition numbers without improving detection, confirming statistical equivalence to unconstrained TS-JEPA within a 0.05 F1 margin. Static extreme-value calibration also fails to maintain its 0.1% design risk under deployment shift; under our exchangeable permutation protocol, inter-block drift accounts for 84% to 96% of this overshoot. Latent residual conditioning suits supervisory monitoring of multi-sensor dynamics, whereas quiet flatlines and baseline ramps require complementary low-variance monitors or adaptive recalibration.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Threshold calibration, Extreme value theory, Representation learning.**

Manuscript received October 15, 2026. This work was supported by the Department of Electrical Engineering and Computer Science at Florida Atlantic University. No prior conference presentation or thesis material is included in this submission. *(Corresponding author: Andre Williams.)*

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I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable digital healthcare [1], [2]. In complex deployments (NASA SMAP, MSL rovers, industrial IoT, and biometric arrays), dozens to hundreds of channels concurrently track power buses, thermal gradients, vibrations, and kinematics [3], where faults must be identified without overwhelming operators with nuisance alerts.

False-Alarm Flooding Under Automated Threshold Calibration: Operational monitoring requires automated decision rules that convert continuous anomaly scores into binary alarms using nominal calibration data alone. Prevailing deep detectors (TimesNet, TranAD, Anomaly Transformer) minimize point-level reconstruction error in raw observation space [4]–[6]. Because raw telemetry contains ambient sensor jitter and channel-scale disparities, observation-space residuals exhibit severe excess kurtosis ($\gamma_2 > 8$) and serial clustering. When paired with automated extreme-value thresholding such as Streaming Peaks-Over-Threshold (SPOT) [7] at nominal design risk $q = 10^{-3}$ (0.1%), these heavy-tailed residuals suffer severe calibration breakdown on unseen nominal data, triggering false alarms across 22%–31% of nominal timesteps.

Latent Residual Conditioning and Scope of Claims: To suppress false-alarm flooding without synthetic anomaly injection [8], we study Joint-Embedding Predictive Architectures (JEPAs) that score predictive discrepancies in a low-dimensional latent space [9], alongside an *Operator-Entropy JEPA* (OpEntropy JEPA) variant regularized by Koopman spectral entropy and a shifted Bregman energy barrier. Physics-inspired regularizers (operator entropy, Reynolds-stress closure, curl-free potential flow) are evaluated strictly as algebraic priors on abstract latent coordinates, not as physical conservation laws, and our experiments separate three distinct operational objectives: (i) suppressing nuisance false alarms under automated calibration, (ii) detecting anomalies reliably across operating points, and (iii) emitting scores early enough to support intervention.

A. Contributions

This study makes four focused contributions:

1) **Operational False-Alarm Reduction vs. Detection Sensitivity:** Across 33 continuous telemetry streams from seven physical datasets (19 non-GHL at five seeds; 14 GHL at three to five seeds), latent predictive scoring via TS-JEPA and OpEntropy JEPA reduces held-out nominal false-positive rates four-fold under retrospective buffered attribution (7.6%–8.6% vs. 30.0% TimesNet and 30.7% TranAD; paired reductions -0.142 to -0.153, $p \leq 0.006$) and $12\times$ –$18\times$ under strictly causal trailing streaming (1.4%–1.9% vs. 22.2%–24.3%). Evaluating both matched calibration quantiles ($q^* \in \{0.001, 0.050\}$) and direct matched realized test-FPR operating points ($f^* \in \{0.01, 0.05, 0.10, 0.20\}$, Fig. 2(b)) across all 19 non-GHL streams shows that latent models improve low-FPR causal event recall (0.439 vs. 0.217 at f^*=0.01), whereas buffered recall, Point-F1, and PR-AUC remain statistically comparable to reconstructive baselines.
2) **Mechanism Isolation (Compression vs. Representation Drift):** Across 12 controlled backbone arms on all 19 non-GHL streams (Section VI-E, Fig. 2(a), Table I(e)), we separate score-distribution compression from representation learning: adding a D=32 LayerNorm bottleneck to an observation forecaster (`TCN-ObsPred-LN`) mechanically reduces calibration kurtosis (γ_2 : 6.08 $\rightarrow$ 1.14 causal) and halves causal FPR (0.246 $\rightarrow$ 0.117), while learned latent projection in TS-JEPA halves calibration-to-holdout Kolmogorov–Smirnov drift (0.386 $\rightarrow$ 0.198, $p = 0.0001$) and lowers causal FPR a further eight-fold (0.117 $\rightarrow$ 0.015).
3) **Operator-Entropy Theory and Factorial Regularizer Null Results:** Proposition 1 distinguishes two regimes: at finite $\epsilon > 0$, the smooth Bregman barrier has vanishing matrix gradient at $\mathbf{K} = \mathbf{0}$ and relies on non-degenerate cross-covariance ($\mathbf{M} \neq \mathbf{0}$) to exclude stationarity at zero, whereas in the $\epsilon \rightarrow 0^+$ limit the barrier alone diverges as $\|\mathbf{K}\|_F \rightarrow 0$. Factorial ablations (Table I(a),(c)) show that spectral regularization drives operator effective rank to 31.97–32.00/32 and condition number to $\kappa(\mathbf{K}) =$ 1.03–1.08, but yields no aggregate detection gain over unconstrained TS-JEPA, confirming Two One-Sided Tests (TOST) equivalence within $\pm$0.05 F1 ($p \leq 0.0054$).
4) **Multi-Risk Calibration Diagnostics, Latency, and Fault Boundaries:** Evaluating SPOT and empirical quantiles across target risks $q \in [10^{-4}, 10^{-1}]$ (Fig. 2(c)) shows that static calibration overshoots nominal design risk across all tested detectors (1.0%–1.5% causal and 5.3%–6.2% buffered FPR at q=10^{-3} for JEPAs vs. 16%–22% for reconstructors), with inter-block drift accounting for 84%–96% of causal overshoot under our exchangeable permutation protocol. Horizon sweeps ($S \in \{1, 4, 16, 32, 64\}$) and fault injections further establish where zero-lag (S=1) monitoring succeeds and where quiet flatline faults (0.02–0.07 recall) require complementary variance monitors.

II. Related Work

Dynamical Systems, Reconstruction, and Forecasting Detectors: Multi-sensor telemetry models system dynamics as state trajectories $\dot{x}(t) = \mathbf{f}(x(t))$ [10]. Neural ODEs [11] parameterize velocity fields in raw observation space, where integrating through high-frequency sensor noise induces numerical stiffness. Koopman theory and dynamic mode decomposition [12]–[14] learn linear operators on observable embeddings, identifying disruptions via spectral decoherence; our proposed Operator-Entropy JEPA architecture transfers this principle to a latent transition operator regularized by spectral entropy.

Reconstructive detectors model nominal patterns via observation reconstruction: LSTM forecasting with dynamic thresholding for spacecraft telemetry [3], association-discrepancy attention in Anomaly Transformer [4], 2D FFT-periodicity variation in TimesNet [5], and adversarial sub-series attention in TranAD [6]. These share a vulnerability: point-level losses bias models toward stochastic noise, accumulating residual variance across irrelevant channels in high-dimensional arrays.

Threshold Calibration, Contrastive, and Self-Supervised Detection: Unsupervised monitoring requires automated rules to map residual scores to binary alarms without labels. Non-parametric thresholding [3] and Streaming Peaks-Over-Threshold (SPOT) [7] fit Generalized Pareto tails under design risk q; however, EVT asymptotic guarantees assume i.i.d. samples, which temporal autocorrelation and operational drift violate.

To avoid reconstructing noise, contrastive methods train discriminative representations: NCAD [8] injects synthetic anomalies, yielding lower calibrated point detection than predictive models (0.058 vs. 0.160 macro Point-F1 across 11 datasets, Table II(d)). DCdetector [15] maximizes dual-attention consistency (0.070 macro Point-F1). Joint-Embedding Predictive Architectures [9], [16], [17] predict target representations in latent space with non-contrastive collapse prevention [18]–[20]. Time-series extensions [21] evaluate ranking or post-hoc threshold sweeps without studying how latent residuals behave under automated tail calibration.

Physics-Inspired Priors and Conceptual Positioning: PINNs [22] and Hamiltonian/Lagrangian networks [23], [24] embed physical conservation laws as rigid differential constraints; their stiff competing gradients [25] and known-PDE requirements fit poorly to uncalibrated multi-sensor telemetry. We strictly distinguish *physics-inspired* algebraic regularizers (where latent coordinates are abstract learned features, not physical state variables) from *physics-informed* conservation constraints; *causal data access* from *score-emission latency*; and *operator non-degeneracy* from *encoder representation diversity* and *anomaly separability*.

III. Multi-Sensor IoT and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams sourced from NASA spacecraft archives [3], the UCI Machine Learning Repository [26], [27], TimeEval [28], and mTSBench [29]: 33 streams are evaluated under the primary multi-seed held-out calibration protocol and 12 reserved for operational case studies. Streams span four physical domains: *spacecraft telemetry* (SMAP and MSL entities, with two short-nominal MSL streams reserved;

SWAN-SF solar magnetograms [30]); *wearable kinematics* (six Daphnet FoG streams [26]: 64 Hz tri-axial accelerometry for Freezing-of-Gait events, so S=64 is a 1-s horizon); *industrial processes* (14 GHL gasoil heating-loop streams, the Genesis pick-and-place testbed, and GECCO water-distribution chemistry/hydrology at 1-minute resolution); and *smart infrastructure* (Room-Occupancy environmental sensing [31], six CICIDS edge network-flow streams, CalIt2 people-flow [27], and Metro I-94 highway traffic).

Each stream provides nominal training sequences and labeled anomaly test sequences; channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatline phases. We evaluate the standardized subsets studied in the literature [3], [28] (of 55 SMAP and 27 MSL archive entities) for strict matching and reproducible runtime.

To eliminate circular in-sample threshold tuning, each stream's nominal sequence is partitioned chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by an unobserved 64-step isolation gap (Section V-A). Stream inclusion in the primary multi-seed held-out protocol was governed by two pre-specified criteria prior to model evaluation: (i) a nominal sequence long enough to yield at least $N_{\text{cal}} \geq 500$ calibration windows after the $65\%/17.5\%/17.5\%$ split, and (ii) tractable stride-1 multi-seed training and evaluation runtime. Exactly 33 streams across seven datasets satisfy both criteria (6 Daphnet, K=9; 14 GHL, K=16; 4 MSL, K=55; 6 SMAP, K=25; Genesis, K=18; Metro, K=5; SWAN, K=38). The remaining 12 streams (CalIt2, K=2; two Room-Occupancy, K=5; two short-nominal MSL; GECCO, K=9; and six 72-channel CICIDS network flows) are evaluated in operational incident case studies and in a broader 11-dataset, 130-stream single-seed baseline cross-validation panel (Table II(d)). The dyadic front-end ablation (Table I(b)) uses a five-stream panel that includes one diagnostic SMD stream (`machine-1-2`, K=38).

Two external-validity caveats bound interpretation. First, SMAP and MSL contain documented mislabeled segments and near-trivial point anomalies [32]; their per-dataset scores represent upper bounds on operational detectability. Second, because multiple streams within SMAP, MSL, Daphnet, and GHL share sensor hardware and environmental telemetry characteristics, streams within a dataset are not independent application domains. We therefore report unweighted Dataset-Macro means, per-dataset between-stream and within-stream seed standard deviations, leave-one-dataset-out (LODO) domain sensitivity, and dataset-clustered permutation tests (Table II(c)), and restrict conclusions strictly to the evaluated entities and sampling regimes rather than extrapolating to entire mission archives.

IV. Methodology: Operator-Entropy Joint-Embedding Predictive Architecture

A. Problem Formulation and Latent Space Prediction

Let $\mathbf{X} \in \mathbb{R}^{T\times K}$ denote a multivariate stream partitioned into context $x_{\text{ctx}} = \mathbf{X}[t-C:t,:] \in \mathbb{R}^{C\times K}$ and suspect horizon $x_{\text{tgt}} = \mathbf{X}[t:t+S,:] \in \mathbb{R}^{S\times K}$. Reconstructive baselines (TimesNet, TranAD, Anomaly Transformer) minimize pointwise error $\min_\theta \mathbb{E}\|\mathbf{X} - \mathcal{D}_\theta(\mathcal{E}_\theta(\mathbf{X}))\|_F^2$, which can weight channels with different noise scales unequally; whether this consumes model capacity is not tested here. We instead adopt the JEPA framework [9]: a context encoder E_θ maps $x_{\text{ctx}} \to z_{\text{ctx}} \in \mathbb{R}^D$ and an EMA momentum target encoder E_ϕ maps $x_{\text{tgt}} \to z_{\text{tgt}}$, with $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$, $m_k \in [0.996, 0.9995]$. The objective forecasts target representations in latent space to filter out stochastic sensor noise and condition prediction residuals for automated tail calibration.

B. Latent Predictor Architectures and Dynamical Inductive Priors

We formulate Operator-Entropy JEPA as our primary architecture, and evaluate it alongside an unconstrained baseline (`TS-JEPA`) and two comparative physical formulations (Reynolds-Stress and Potential-Flow):

1) Operator-Entropy JEPA (Spectral Dynamics & Modal Dispersion): Motivated by Koopman operator theory [33], [34], where nonlinear trajectories are mapped via linear operators on observable functions, we parameterize the predictive latent transition via a learned operator $\mathbf{K} \in \mathbb{R}^{D\times D}$: $\hat{z}_{\text{tgt}} = \mathbf{K}z_{\text{ctx}}$. To prevent collapse onto a low-rank subspace (dimensional collapse) or to zero (energy collapse), the regularizer decouples spectral uniformity from total operator energy.

Let $\sigma_1 \geq \cdots \geq \sigma_D \geq 0$ denote singular values of $\mathbf{K}$, $\lambda_i = \sigma_i^2$ eigenvalues of $\mathbf{K}\mathbf{K}^\top$, and $\mathcal{E}(\mathbf{K}) = \|\mathbf{K}\|_F^2 = \sum_i \lambda_i$ total energy. To ensure collapsing modes ($\lambda_i \to 0^+$) receive strictly negative gradients ($\partial\Delta\mathcal{H}/\partial\lambda_i < 0$) without sign reversal, we form the smoothed modal distribution $p_i = (\lambda_i + \epsilon_{\log})/\sum_{j=1}^D(\lambda_j + \epsilon_{\log})$ ($\epsilon_{\log} = 10^{-12}$) and Von Neumann spectral entropy $\mathcal{H}_{\text{spectral}}(\mathbf{K}) = -\sum_{i=1}^D p_i \log p_i$. The *Spectral Entropy Deficiency* measures Kullback-Leibler divergence from the uniform equipartition $\mathbf{u} = [1/D, \ldots, 1/D]^\top$:

$$\Delta\mathcal{H}(\mathbf{K}) = \log D - \mathcal{H}_{\text{spectral}}(\mathbf{K}) = \mathcal{D}_{\text{KL}}(\mathbf{p} \,\|\, \mathbf{u}) \geq 0, \quad (1)$$

where $\Delta\mathcal{H} = 0$ if and only if all modes carry equal energy ($\sigma_1 = \cdots = \sigma_D$), while rank-1 collapse yields $\Delta\mathcal{H} \approx \log D$. Minimizing $\Delta\mathcal{H}$ drives $\mathbf{K}$ toward isometric modal equipartition rather than low-rank truncation, acting directly on learned embeddings.

To penalize energy collapse ($\|\mathbf{K}\|_F \to 0$) without explosive gain or zero-gradient clamping, we impose a smooth shifted Bregman energy barrier relative to nominal isometric scale $\mathcal{E}_0 = \gamma D$ ($\gamma = 1.0$):

$$\Phi_{\text{energy}}(\mathbf{K}) = r(\mathbf{K}) - 1 - \log\big(r(\mathbf{K}) + \epsilon\big) + \log(1+\epsilon) \geq 0, \quad (2)$$

where $r(\mathbf{K}) = \|\mathbf{K}\|_F^2/(\gamma D)$ and $\epsilon = 10^{-6}$. This barrier is C^∞ smooth for $r \geq 0$, vanishes at $r = 1$, caps zero-energy penalty at $\approx 12.8\,\beta_{\text{norm}}$, and has radial derivative $\partial\Phi_{\text{energy}}/\partial r|_{r=0} = 1 - 1/\epsilon < 0$. With $\mathbf{K} \sim \mathbf{I} + \mathcal{N}(\mathbf{0}, 0.01^2\mathbf{I})$, $r \approx 1$ and initial barrier penalty vanishes. The composite regularizer is:

$$\mathcal{R}_{\text{spectral}}(\mathbf{K}) = \Delta\mathcal{H}(\mathbf{K}) + \beta_{\text{norm}}\Phi_{\text{energy}}(\mathbf{K}), \quad (3)$$

and the training objective minimizes $\mathcal{L}_{\text{Op-Entropy}} = \frac{1}{BD}\sum_b \|\mathbf{K}z_{\text{ctx}}^{(b)} - z_{\text{tgt}}^{(b)}\|_2^2 + \lambda_{\text{ent}}\mathcal{R}_{\text{spectral}}(\mathbf{K}) + \beta_{\text{Lyap}}\mathcal{L}_{\text{Lyap}}$ ($\beta_{\text{norm}} = 1.0, \lambda_{\text{ent}} = 0.5, \beta_{\text{Lyap}} = 0.02$). The optional discrete Lyapunov defect aligns batch covariances, $\mathcal{L}_{\text{Lyap}} = \|\boldsymbol{\Sigma}_{\text{pred}} - \boldsymbol{\Sigma}_{\text{tgt}}\|_F^2/\big(D\,(\text{tr}\,\boldsymbol{\Sigma}_{\text{tgt}}/D + 10^{-6})^2\big)$, where $\boldsymbol{\Sigma}_{\text{pred}}, \boldsymbol{\Sigma}_{\text{tgt}}$ are the empirical covariances of $\mathbf{K}z_{\text{ctx}}$ and z_{tgt} over the batch. Shared representation stages are fixed across

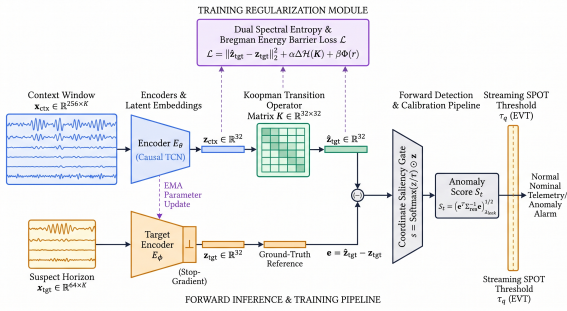


Fig. 1. Operator-Entropy JEPA architecture: multiscale context ($\mathbf{x}_{\text{ctx}} \in \mathbb{R}^{256\times K}$) and suspect ($\mathbf{x}_{\text{tgt}} \in \mathbb{R}^{64\times K}$) windows are mapped into latent space via online (E_θ) and EMA momentum (E_ϕ) encoders with a stop-gradient barrier ($\perp$). A linear Koopman transition operator ($\mathbf{K} \in \mathbb{R}^{32\times 32}$) forecasts the target representation $\hat{\mathbf{z}}_{\text{tgt}} = \mathbf{K}\mathbf{z}_{\text{ctx}}$, regularized by dual Von Neumann spectral entropy and a logarithmic Bregman energy barrier against collapse. Residual discrepancies pass through a coordinate saliency gate to yield anomaly score S_t calibrated via streaming SPOT.

controls. $\mathcal{R}_{\text{spectral}}(\mathbf{K})$ acts on $\mathbf{K}$ only; encoder collapse is mitigated empirically by EMA target dynamics and stop-gradient operations [16], [35].

Proposition 1 (Operator Non-Triviality Under Finite-ϵ and Singular Barriers)**.** *Let* $\mathcal{L}_\epsilon(\mathbf{K}) = \frac{1}{BD}\sum_{b=1}^{B}\|\mathbf{K}z_{ctx}^{(b)} - z_{tgt}^{(b)}\|_2^2 + \lambda_{ent}\mathcal{R}_{spectral}(\mathbf{K})$ *with* $\lambda_{ent}, \beta_{norm} > 0$*, target batch energy* $E_{tgt} > 0$*, and empirical cross-covariance* $\mathbf{M} = \frac{1}{BD}\sum_{b=1}^{B} z_{tgt}^{(b)} z_{ctx}^{(b)\top}$*: (i)* Finite-$\epsilon$ smooth regime ($\epsilon = 10^{-6}$): $\Phi_{energy}(\mathbf{0}) \approx 12.8$ *is finite and* $\partial\Phi_{energy}/\partial r|_{r=0} < 0$*, but the matrix gradient* $\nabla_{\mathbf{K}}\mathcal{R}_{spectral}(\mathbf{0}) = \mathbf{0}$ *vanishes because* $r(\mathbf{K}) \propto \|\mathbf{K}\|_F^2$*. The barrier alone produces no repulsive gradient at* $\mathbf{K} = \mathbf{0}$*; instead, whenever* $\mathbf{M} \neq \mathbf{0}$*,* $\nabla_{\mathbf{K}}\mathcal{L}_\epsilon(\mathbf{0}) = -2\mathbf{M} \neq \mathbf{0}$ *excludes* $\mathbf{K} = \mathbf{0}$ *as a stationary point or minimizer. (ii)* Singular-barrier limit ($\epsilon \to 0^+$): *For unshifted* $\mathcal{L}_0(\mathbf{K})$*,* $\lim_{\|\mathbf{K}\|_F\to 0}\Phi_{energy}(\mathbf{K}) = +\infty$ *and* $\lim_{\|\mathbf{K}\|_F\to 0}\mathcal{L}_0(\mathbf{K}) = +\infty$ *unconditionally, independent of* $\mathbf{M}$*.*

Proof. (i) For $\epsilon > 0$, Φ_{energy} and $\Delta\mathcal{H}$ are smooth. By the chain rule, $\nabla_{\mathbf{K}}\Phi_{\text{energy}}(\mathbf{K}) = \frac{2}{\gamma D}\Phi'_{\text{energy}}(r)\,\mathbf{K} = \mathbf{0}$ at $\mathbf{K} = \mathbf{0}$. At $\mathbf{K} = \mathbf{0}$, $\boldsymbol{\rho}(\mathbf{0}) = \mathbf{I}/D$ minimizes the nonnegative smooth entropy deficiency $\Delta\mathcal{H}$, so $\nabla_{\mathbf{K}}\Delta\mathcal{H}(\mathbf{0}) = \mathbf{0}$. Differentiating the prediction loss gives $\nabla_{\mathbf{K}}\mathcal{L}_\epsilon(\mathbf{0}) = -2\mathbf{M}$; if $\mathbf{M} \neq \mathbf{0}$, $\mathbf{K} = \mathbf{0}$ is non-stationary. (ii) For $\epsilon = 0$, $\Phi_{\text{energy}} = r - 1 - \log r \to +\infty$ as $r \to 0^+$; nonnegativity of the other terms yields $\lim_{\|\mathbf{K}\|_F\to 0}\mathcal{L}_0(\mathbf{K}) = +\infty$. □

Remark (Scope, Cross-Covariance Verification, and Non-Collapse): Proposition 1 guarantees operator non-degeneracy, not representation diversity or anomaly detection. Unlike representation regularizers (VICReg [20], Barlow Twins [18], whitening [19]) that penalize covariance collapse of encoder outputs $z_{\text{ctx}}, z_{\text{tgt}}$, $\mathcal{R}_{\text{spectral}}(\mathbf{K})$ constrains only the transition spectrum. If E_θ maps all inputs to a constant, $\mathbf{K} = \mathbf{I}$ attains zero loss; conversely, an isometric operator ($\kappa(\mathbf{K}) \approx 1$) is compatible with representations in a low-dimensional subspace ($\text{RankMe}(z_{\text{ctx}})\approx 5.2$, $\text{RankMe}(z_{\text{tgt}})\approx 10.0$ of 32; Table I(c)). Empirically, target variance ($E_{\text{tgt}} \in [0.09, 5.88]$) and coupling ($\|\mathbf{M}\|_F > 0$) hold across all 195 primary runs because adjacent windows share LayerNorm-standardized features under cosine EMA $m_k \in [0.996, 0.9995]$ [35].

2) Auxiliary Quadratic Second-Moment Head: Inspired by Reynolds decomposition in fluid turbulence, which separates mean flow from velocity fluctuations, we evaluate an auxiliary quadratic second-moment head in latent space. The observed matrix $\boldsymbol{\Sigma}_{\text{obs}} = (\mathbf{P}z_{\text{tgt}})(\mathbf{P}z_{\text{tgt}})^\top + 10^{-3}\mathbf{I} \in \mathbb{S}_{++}^{d_s\times d_s}$ ($d_s = 16$, via projection $\mathbf{P} \in \mathbb{R}^{d_s\times D}$) is an instantaneous rank-1 projected outer product plus ridge penalty, not a physical turbulent covariance. A closure head outputs lower-triangular $\mathbf{L} \in \mathbb{R}^{d_s\times d_s}$ reconstructing predicted covariance $\boldsymbol{\Sigma}_{\text{pred}} = \mathbf{L}\mathbf{L}^\top + 10^{-3}\mathbf{I}$. The objective minimizes $\mathcal{L}_{\text{Reynolds}} = \text{MSE}(\hat{z}_{\text{tgt}}, z_{\text{tgt}}) + 0.1\ \text{MSE}(\boldsymbol{\Sigma}_{\text{pred}}, \boldsymbol{\Sigma}_{\text{obs}})$. At test time, scoring adds the second-moment discrepancy: $\mathcal{S}(t) = g(\hat{z}_{\text{tgt}} - z_{\text{tgt}}) + 0.5\,g(\text{vec}(\boldsymbol{\Sigma}_{\text{obs}} - \boldsymbol{\Sigma}_{\text{pred}}))$.

3) Potential-Flow JEPA (Conservative Field Theory and Flow Matching): Under Helmholtz-Hodge decomposition a smooth velocity field splits into a curl-free gradient and a solenoidal part. We parameterize $\hat{v} = -\nabla_{z_t}\Phi_\psi$ via a residual MLP conditioned on state z_t, flow time $t \in [0,1]$, context z_{ctx}, and a regime projection selected among $K_r = 4$ orthonormal subspaces (d=8, D=32) with tangent projection $v_\perp = v - \langle u, v\rangle u$, $u = z_t/\|z_t\|_2$. Training uses conditional flow matching along linear paths from noise $z_{\text{noise}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to z_{tgt}, $\mathcal{L}_{\text{Pot-Flow}} = \mathbb{E}_{t, z_{\text{noise}}}\|\hat{v} - (z_{\text{tgt}} - z_{\text{noise}})\|_2^2$, and inference queries the field at the trajectory midpoint, $\mathbf{e}(t) = \hat{v}(0.5z_{\text{tgt}}, 0.5, z_{\text{ctx}}, p_{\text{regime}}) - z_{\text{tgt}}$. The curl-free bias is hypothesized to penalize cyclic nominal orbits and degrade detection on periodic kinematics (Section VII-B).

C. Anomaly Scoring and Operational Detection Latency

During inference, windows $x_{\text{ctx}} = \mathbf{X}[t - C : t, :]$ and $x_{\text{tgt}} = \mathbf{X}[t : t + S, :]$ are processed at stride 1. For TS-JEPA and Potential-Flow, the score gates the residual $\mathbf{e}(t) = z_{\text{tgt}}(t) - \hat{z}_{\text{tgt}}(t)$ (or whitened $\boldsymbol{\Omega}^{1/2}(\mathbf{e}(t) - \boldsymbol{\mu}_r)$) via $g(\cdot)$ (Section IV-D1): $\mathcal{S}(t) = g(\mathbf{e}(t))$. Because final LayerNorm projects embeddings to radius $\sqrt{D}$, Operator-Entropy JEPA tracks pre-LayerNorm radial energy scaled by window RMS $(e_{\text{ctx}}, e_{\text{tgt}})$ with $e_{\text{pred}} = e_{\text{ctx}}\|\hat{z}_{\text{tgt}}\|_2/(\|z_{\text{ctx}}\|_2 + 10^{-6})$, scoring $\mathcal{S}(t) = g(\mathbf{e}(t)) + 0.2\,|e_{\text{tgt}}(t) - e_{\text{pred}}(t)|/(e_{\text{pred}}(t) + 10^{-6})$. For Reynolds-Stress, $\mathcal{S}(t) = g(\mathbf{e}(t)) + 0.5\,g(\text{vec}(\boldsymbol{\Sigma}_{\text{obs}} - \boldsymbol{\Sigma}_{\text{pred}}))$. Shared stages are fixed across controls (Section VI-A).

Three-Part Operational Causality Framework: To resolve the methodological confounding between data availability, decision latency, and evaluation attribution, we formalize detection under three operational criteria:

(a) **Data causality ($\mathcal{F}_\tau$-measurability):** At decision time $\tau = t + S - 1$, the model observes only history

$\mathbf{X}_{:\tau} = \mathbf{X}[t-C:\tau,:]$. Neither encoder accesses future observations beyond τ.

(b) **Score-emission latency ($S-1=63$ steps):** Because suspect representation z_{tgt} encodes the horizon $\mathbf{X}[t:t+S,:]$, window score s_t cannot be computed until timestep $\tau = t+S-1$. This imposes an inherent emission latency of $S-1=63$ steps: 0.98 s on Daphnet (64 Hz), 10.5 min on SMAP, MSL, and GHL under standard 10-s (0.1 Hz) sampling [3], [29], 63 min on GECCO (1 sample/min), and 63 h on Metro (1 sample/h). This latency guarantees zero future lookahead, but must not be confused with instantaneous detection at fault inception: batch scoring over horizon S=64 fits supervisory health monitoring and slow subsystem drift where anomaly persistence distinguishes faults from noise, but cannot provide subsecond emergency tripwires on low-frequency telemetry. Section VII-D quantifies how shortening S trades this latency against false-alarm control.

(c) **Alarm attribution convention:** We distinguish two attribution regimes: (1) *strictly causal trailing attribution*, where score s_t is emitted and evaluated strictly at τ, providing zero-lookahead online monitoring; and (2) *retrospective buffered smear attribution*, where s_t is attributed across the entire suspect window $[t+1, t+S]$. Under unadjusted point evaluation, buffered alarms in $[t^*-S, t^*]$ register as false alarms: buffering provides fixed-lag detection, not early warning. The two regimes represent distinct operational realities and are reported separately throughout (Table II(a),(b)).

D. Shared Internal Inductive Biases

To stabilize learning across high-dimensional, noisy telemetry, the latent JEPA architectures incorporate shared inductive mechanisms:

1) Coordinate Saliency Gating and the Auxiliary Channel-Gate Variant: On high-dimensional streams ($K \geq 25$), faults frequently originate in a localized subsystem while the remaining channels exhibit ambient noise, diluting localized anomalies in isotropic representations. The deployed detector mitigates this in latent space via an anti-drowning coordinate saliency gate $g(\mathbf{e}) \equiv \mathcal{D}_{\text{sal}}(\mathbf{e})$ on error vectors $\mathbf{e} = \hat{z}_{\text{tgt}} - z_{\text{tgt}} \in \mathbb{R}^D$:

$$\mathcal{D}_{\text{sal}}(\mathbf{e}) = (1-\alpha)\|\mathbf{e}\|_2 + \alpha\sqrt{D\sum_{d=1}^{D} w_d e_d^2}, \qquad (4)$$

$$w_d = \frac{\exp(|e_d|/\tau)}{\sum_{j=1}^{D}\exp(|e_j|/\tau)},$$

amplifying sparse coordinate deviations by up to $\sqrt{D}$. We fix blend parameter $\alpha = 0.5$ and softmax temperature $\tau = 0.5$ across all latent predictors. Reconstructive baselines (TimesNet, TranAD) evaluate unweighted observation errors and do not use the gate. This is the discrepancy gate used in every latent experiment: input channels are encoded unweighted, and the gate acts only on the D-dimensional predictive residual.

a) Auxiliary Input-Channel Gate (Evaluated Variant): A strictly causal gate weights each channel by the sigmoid of its standardized context-window variance share, $g_k = \sigma((s_k - \bar{s})/(\sigma_s + \epsilon))$, applied symmetrically to both encoders so that $\partial g_k/\partial \mathbf{x}_{\text{tgt}} \equiv \mathbf{0}$ (identity at K=1). It cannot isolate quiet faults (recall 0.02–0.07 under every gating regime, Section VI); the deployed pipeline weights inputs uniformly.

2) Multiscale Frequency Front-End: Anomalies span temporal scales from sensor spikes to thermal drifts; the encoder front-end therefore splits each channel into J=4 smooth dyadic shells on an $N_{\text{fft}} = 256$ rFFT grid, normalized bin-wise into an exact amplitude partition of unity, $\sum_j \Psi_j[\omega_m] \equiv 1$ (measured deviation $< 1.2\times10^{-7}$). Each shell feeds a causal 1D convolution ($k \in \{15, 9, 5, 3\}$ coarse-to-fine), recombined by learned scale routing. Because the shells overlap smoothly at band edges, the front-end is *less* frequency-localized than standard wavelet banks. We therefore describe it as a Fourier-domain multiscale front-end, not an exact Littlewood–Paley octave decomposition, and measure its empirical contribution by matched ablation (Table I(b)).

3) Kurtosis-Adaptive Extreme Value Theory (EVT) and Calibration Realities: Binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_t)$ are generated via Streaming Peaks-Over-Threshold (SPOT) [7]: excesses over a high threshold u (98th percentile) asymptotically follow a Generalized Pareto Distribution (GPD) with scale σ and shape ξ:

$$\tau = u + \frac{\sigma}{\xi}\left[\left(\frac{qN}{N_u}\right)^{-\xi} - 1\right], \qquad (5)$$

where $q = 10^{-3}$ denotes the design risk probability, N is total calibration samples, and N_u is the count of excesses. To account for varying tail heaviness across diverse sensor regimes, we evaluate an empirical heuristic multiplier $\tau_{\text{eff}} = \tau\cdot[1+\frac{1}{6}\tanh(\gamma_2)]$ governed by the sample excess kurtosis γ_2 of calibration residuals: on heavy-tailed distributions ($\gamma_2 > 0$), τ_{eff} rises to suppress volatile noise; on light-tailed ones, it tightens for sensitivity.

The multiplier is an engineering heuristic for pre-asymptotic tails, not an EVT theorem. The primary pipeline uses standard SPOT ($u = p_{98}, q = 10^{-3}$) and evaluates the multiplier only as a sensitivity arm (Section VII-F), where it degrades out-of-sample risk control (5.26% → 17.68% FPR).

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup and Hyperparameters

To ensure rigorous and fair comparison, all architectures are trained and evaluated under an identical streaming time-series pipeline. Telemetry streams are processed using a sliding window with past context duration $C = 256$ and suspect target horizon $S = 64$ (total window length $L = 320$), evaluated with a stride of 1 timestep across all testing streams.

Context representations $z_{\text{ctx}} \in \mathbb{R}^D$ ($D = 32$) are extracted via a multiscale dyadic front-end and causal dilated TCN (6 residual blocks, F=48, kernel 5, dilations $\{1, 2, 4, 8, 16, 32\}$); multi-statistic pooling (terminal, mean, max, and channel statistics) projects the C=256 context through a linear bottleneck recording pre-LayerNorm radial energy before LayerNorm and dropout (p=0.20). The momentum target encoder uses cosine EMA m_{base}=0.996 → $m_{\max}$=0.9995.

Models are optimized using AdamW with an initial learning rate $\eta = 10^{-3}$ decayed to $\eta_{\min} = 10^{-5}$ via Cosine Annealing, weight decay 10^{-4}, and mini-batch size $B = 32$ across 20 training epochs on nominal telemetry sequences.

a) Held-Out Calibration Protocol: To evaluate out-of-sample threshold calibration without circularity, each stream's nominal telemetry is partitioned into three disjoint chronological blocks: *training* (first 65%), *nominal calibration* (next 17.5%), and *nominal evaluation* (final 17.5%, separated by an unobserved 64-step isolation gap). The training block optimizes representation weights alone. Withheld from training, the nominal calibration block fits per-stream GPD tail parameters σ, ξ above u=0.98 at q=10^{-3}, estimating γ_2 strictly within this interval. The nominal evaluation block then measures held-out FPR on telemetry unobserved during threshold selection. Thresholds remain fixed during test evaluation to measure static EVT stability under drift; dynamic rolling SPOT is evaluated separately in the operational incident analysis (Section VII-C).

b) Calibrator Assignment: All experiments use standard SPOT ($u = p_{98}$, $q = 10^{-3}$, no kurtosis multiplier) on held-out nominal splits, except the rolling-SPOT incident analysis and the calibrator comparison itself (Table IV); no table mixes calibrators across models.

c) Multi-Seed Replication, Primary Endpoints, and Multiplicity Controls: Principal models on the 19 non-GHL streams are trained across five seeds (42, 123, 456, 7, 2024); the 14 large GHL streams ($> 10^6$ steps) use three seeds (five on the two smallest) for cost, and ablation controls use three. We report held-out nominal FPR, Point-F1, and PR-AUC as distinct outcomes; nonsignificant detection differences are not evidence of parity.

Inference reports stream-macro means, seed SDs, and two-sided Wilcoxon signed-rank tests. Within each stream, per-cell FPR intervals use a moving-block bootstrap (L=64) over the false-alarm indicator sequence; across streams, paired contrasts resample stream differences (2,000 iterations, seeds averaged within stream). The independent statistical unit is the telemetry stream (n=19 non-GHL at five seeds; n=33 where GHL enters at available coverage), with dataset-clustered checks reported to account for shared sensor environments. For general detector contrasts, nonsignificant tests do not establish equivalence; instead, we report paired differences with effect sizes and CIs. Across general contrasts, p-values are Benjamini–Hochberg FDR-adjusted at $q_{\text{FDR}} = 0.05$; CIs spanning zero or adjusted $p > 0.05$ are treated as inconclusive. For physical regularizer variants, where hypotheses test whether additions alter baseline performance, we complement Wilcoxon tests with formal Two One-Sided Tests (TOST) against TS-JEPA: the null hypothesis $H_0 : |\mu_1 - \mu_2| \geq \Delta$ tests non-equivalence against $H_1 : |\mu_1 - \mu_2| < \Delta$ under an equivalence margin of $\Delta = \pm 0.05$ Point-F1. Every reported test is stated in the text with its endpoint, cohort, number of paired streams, raw and adjusted p, and effect-size interval.

B. Baseline Architectures and Parameter Capacities

To isolate algorithmic principles from implementation-specific discrepancies, all baselines are executed within a unified PyTorch benchmarking pipeline with identical windowing ($C = 256, S = 64$), standardization, and EVT calibration. TimesNet and TranAD are in-house re-implementations configured to published architectural depths and evaluated under a shared optimization budget (AdamW, $\eta = 10^{-3}$, cosine decay, 20 epochs); no method received per-model hyperparameter tuning beyond the shared configuration, restricting claims to these tested implementations. Three checks bound this protocol: (i) *Training-budget sensitivity* (7 streams, 3 seeds, held-out SPOT): halving training to 10 epochs shifts JEPA vs. reconstructor Point-F1 from 0.048 vs. 0.154 to 0.038 vs. 0.158 and held-out FPR from 0.092 vs. 0.333 to 0.087 vs. 0.333. (ii) *Native windows*: at native horizons (TranAD W=10, TimesNet W=96), heavy tails persist ($\gamma_2 = 448.25$ and 98.14, TimesNet FPR 44.09%) vs. $\gamma_2 = 3.14$ for TS-JEPA. (iii) *Published-score replication*: baselines replicate published PA-F1 on SMD and MSL, and Table II(d) extends comparisons to NCAD (149,634 parameters at K=25), DCdetector (298,400), and Anomaly Transformer (4,785,120) on 130 streams.

Compared Models and Reproducibility Specification: Operator-Entropy JEPA (285,701 parameters at K=25, -1.16% vs. unconstrained TS-JEPA at 289,061); physical controls Reynolds-Stress (296,037, $+2.4\%$) and Potential-Flow (338,209, $+17.0\%$); observation baselines TimesNet [5] (648,553) and TranAD [6] (205,746); Linear AR Forecaster (λ_{ridge}=1.0); and 12 controlled mechanism arms (Section VI-E). Non-finite values are replaced by zero after centering ($\boldsymbol{\sigma} \geq 10^{-2}$). SPOT fits GPD parameters (ξ, σ) above $u = p_{98}$ via Grimshaw's procedure with L-BFGS-B fallback ($\xi \in [-0.9, 10], \sigma \geq 10^{-6}$); when $N_u < 5$ or $|\xi| < 10^{-6}$, it falls back to exponential tail extrapolation or the $(1 - q)$ empirical quantile. All 195 primary runs converged without restarts ($\min \text{SD}(s_{\text{cal}}) = 0.0043$) on PyTorch 2.4 / CUDA 12.4 on an RTX 5090 GPU (0.60–0.84 ms/window, 183.8–185.3 MB peak VRAM for JEPAs).

C. Multi-Tier Statistical Evaluation Metrics

Critical surveys [28], [36] document severe reporting vulnerabilities in this literature (point-adjustment inflation, channel-averaging imbalances, threshold gaming), so we report metrics across complementary tiers. *Exploratory operating characteristics*: event recall over nominal calibration budgets and over a realized-test-FPR interval $[0.10, 0.50]$, interpolated within stream–seed curves only where both bounds are observed. *Secondary endpoints*: Dataset-Macro Point-F1 (unweighted mean across datasets, preventing the 14-stream GHL block from dominating); channel-macro Point-F1; PR-AUC (average precision, ranking quality independent of calibration); event recall (fraction of contiguous anomaly segments containing ≥ 1 alarm); mean detection delay Δt (onset-to-first-alarm lag over detected events only); and the trivial all-positive reference $\text{F1}_{\text{all-pos}} = 2\pi/(1 + \pi)$ at base rate π; models below it underperform a constant alarm. Pooled micro-F1 exposes flood-prone models; contiguous PA-F1 and test-sweep PA-F1 quantify the residual calibration gap.

VI. Ablation Studies and Analysis

A. Ablation 1: Physical Regularizer Controls on Fixed Architectures

To test whether the physics-inspired regularizers benefit detection, Table I(a) reports controlled ablations across all 19 non-GHL streams (principal models at five seeds, controls at three, under the held-out protocol), disabling each physical term on a fixed backbone. Because controls run three seeds against the five-seed principal runs, the paired control means below are computed on the shared three-seed subset and differ slightly from the five-seed Table I(a) rows:

1) **Operator-Entropy vs. $\lambda_{\text{ent}} = 0$:** Disabling the spectral regularizer leaves Point-F1 unchanged (0.241 ± 0.014 vs. 0.244 ± 0.024; difference -0.003, 95% interval $[-0.011, +0.005]$, Wilcoxon $p = 0.62$); PR-AUC stays 0.347–0.348.
2) **Reynolds-Stress vs. $\alpha_{\text{stress}} = 0$:** Disabling the stress loss yields 0.203 ± 0.059 vs. 0.208 ± 0.071 ($+0.005$, $[-0.006, +0.016]$, $p = 0.41$).
3) **Potential-Flow vs. no-regime (K_r=1):** Disabling regime routing shifts Point-F1 to 0.129 from 0.096 on the shared three-seed subset ($p_{\text{raw}} = 0.048, p_{\text{BH}} = 0.24$; 0.089 at five seeds in Table I(a)); both underperform TS-JEPA (0.234, $p_{\text{raw}} = 0.013, p_{\text{BH}} = 0.24$, non-significant under FDR adjustment).

Across the full 19-stream suite, the physical regularizers produce no statistically detectable improvement over unconstrained latent prediction. Formal Two One-Sided Tests (TOST) reject non-equivalence ($H_0 : |\Delta| \geq 0.05$) and confirm statistical equivalence to TS-JEPA within a tight $\Delta = \pm 0.05$ Point-F1 margin: Operator-Entropy achieves $p = 0.0009$ (95% CI $[-0.014, +0.021]$, also passing $\Delta = \pm 0.03$ with $p = 0.0283$) and Reynolds-Stress achieves $p = 0.0054$ (95% CI $[-0.042, +0.007]$). These null results establish practical equivalence rather than an empirical advantage.

B. Ablation 2: Factorial Isolation of the Entropy Term and Energy Barrier

A single on/off control cannot separate the spectral-entropy term from the energy barrier or show how either scales. Table I(c) crosses the two terms (neither, entropy only, barrier only, both) and sweeps $\lambda_{\text{ent}} \in \{0.05, 0.10, 0.50, 1.00, 2.00\}$ with the barrier on, on six streams at three seeds with a D=32 operator. Each arm reports operator rank $\mathrm{RankMe}(\mathbf{K})$, condition number $\kappa(\mathbf{K})$, representation rank $\mathrm{RankMe}(z_{\text{ctx}})/\,\mathrm{RankMe}(z_{\text{tgt}})$, the coefficient of variation $\mathrm{CV}_{\mathcal{L}}$ of the training loss as a stability proxy, and held-out calibration and detection outcomes in both attribution regimes.

Three observations follow. First, the unconstrained operator is already near-isotropic ($\mathrm{RankMe}(\mathbf{K})$=31.83 of 32, κ=1.235) because $\mathbf{K}$ is initialized at the identity; the regularizers tighten it only marginally (31.97 and κ=1.083 at the deployed λ_{ent}=0.50, κ=1.034 at λ_{ent}=2.00) and lower $\mathrm{CV}_{\mathcal{L}}$ from 1.013 to 0.917. Second, the encoder representation is unaffected: $\mathrm{RankMe}(z_{\text{ctx}})$ stays within 5.15–5.34 and $\mathrm{RankMe}(z_{\text{tgt}})$ within 9.92–10.07 of 32 in every arm. Operator non-degeneracy, representation diversity, and anomaly separability are therefore separate properties, and only the first moves. Third, no arm or strength improves detection: buffered Point-F1 spans 0.076–0.094 and causal Point-F1 0.022–0.028 with no monotone trend in λ_{ent}, buffered FPR spans 0.066–0.104, and the unconstrained arm has the lowest buffered FPR of all arms (0.066) and the second-highest causal event recall (0.516, behind 0.544 at λ_{ent}=1.00). The entropy term is a well-posed constraint that preserves operator conditioning, not a demonstrated source of detection or calibration gain.

Decoupling. On eight streams (causal trailing), adding covariance coupling (α_Z=0.005) and the Lyapunov defect (β_{Lyap}=0.02) cut delay from 112.0 (single-operator OpEntropy) to 30.1 steps at event recall 0.433 and FPR 0.021, against 85.1, 0.376, and 0.018 for unconstrained TS-JEPA, at some cost in Point-F1 (0.050 vs. 0.060–0.062). These exploratory contrasts shift the recall-delay trade-off and are not part of the headline results.

C. Ablation 3: Auxiliary Input-Channel Gating and Quiet-Fault Reference Monitors

The auxiliary causal input-channel gate is compared against uniform pooling, a non-causal target-only leaky control, and independent context/target gating under an identical TS-JEPA configuration (seed 42, four streams). Macro Point-F1 ranks uniform weighting (0.118) above context gating (0.077), with the leaky control at 0.128 and independent gating weakest (0.071); the effect is stream-dependent in sign (MSL `M-1`: 0.047 → 0.227; SMAP `P-3`: 0.280 → 0.019). The gate is an evaluated variant; the deployed pipeline retains uniform input weighting.

Table I(d) injects six single-channel fault classes into six multivariate streams. The variance gate roughly doubles point recall relative to uniform pooling (macro 0.045 vs. 0.023) at about 3.6× the nominal FPR (0.069 vs. 0.019), and a median-absolute-deviation variant is marginally stronger (0.060 recall). Quiet faults (flatline, stuck value, dropout) stay weakly detected under every latent gating regime (point recall 0.02–0.07). Two observation-space reference monitors test whether a complementary score can cover this blind spot: a rolling channel-variance floor (`lowvar`) and a maximum channel z-deviation (`dev_maxz`). The variance floor reaches point recall 0.18 (0.32 on mid-variance flatlines) and the z-deviation monitor 0.09–0.11, but at nominal FPR of 0.14–0.16, roughly eight times that of the latent detector. A complementary variance monitor recovers part of the quiet-fault gap only at a substantial alarm cost, so quiet-fault coverage remains an open limitation.

D. Ablation 4: Multiscale Dyadic-Shell Front-End

Replacing the standard front-end with the Fourier-domain dyadic shells plus per-shell causal convolutions and learned routing yields a modest, stream-dependent effect (Table I(b)): macro buffered Point-F1 improves from 0.088 to 0.131 across a five-stream exploratory panel (including two diagnostic streams, SMD and GECCO, outside the 33 primary streams),

concentrated on MSL and SMD, with a small SMAP regression. The block adds $\approx 1.6 \times 10^5$ parameters and shell overlap prevents octave isolation, so it is a modest contributor, not the primary driver.

E. Mechanism-Discrimination Study: Latent Space vs. Observation Space

To isolate the operational mechanisms governing tail conditioning and threshold transfer, all 19 non-GHL streams were evaluated under the primary held-out protocol (five seeds, identical windows, AdamW budget, and SPOT u=0.98/q=10^{-3} calibration) across 12 controlled backbone arms (Table I(e), Fig. 2(a); neural variants sharing the `HybridTCNEncoder` with C=256, S=64, d=32, alongside linear and non-parametric controls).

Decoupling Score Compression from Representation Learning: Contrasting these controlled backbones (Table I(e) and Fig. 2(a)) separates score-distribution compression from learned representation dynamics: (1) *Unnormalized observation forecasting (`TCN-ObsPred`):* In raw observation space, residual excursions exhibit severe causal tail kurtosis ($\gamma_2 \approx 6.08$), pronounced calibration-to-holdout Kolmogorov–Smirnov distribution drift ($D_{\text{KS}} = 0.460$), and severe causal nominal false alarms (0.246). (2) *Observation forecasting with LayerNorm (`TCN-ObsPred-LN`):* Introducing a D=32 LayerNorm bottleneck mechanically curtails residual excursions, compressing causal kurtosis to $\gamma_2 \approx 1.14$ and halving causal FPR to 0.117 (buffered 0.186). However, calibration-to-holdout drift persists at $D_{\text{KS}} = 0.386$. Bounded geometry alone halves false alarms through score compression, but cannot resolve distribution drift across temporal blocks. (3) *Latent projection and representation drift suppression:* Shifting prediction into learned latent coordinates fundamentally alters drift. Unconstrained latent prediction (`TS-JEPA`) halves calibration-to-holdout KS drift ($0.386 \rightarrow 0.198$, paired Wilcoxon $p = 0.0001$, $\gamma_2 = 1.18$) and suppresses causal false alarms an additional eight-fold ($0.117 \rightarrow 0.015$, event recall 0.471). (4) *Foresight vs. geometric compression:* Autoencoding context representations without temporal foresight (`JEPA-CtxRecon`) matches this drift reduction ($D_{\text{KS}} = 0.199$, $\gamma_2 = 2.49$) and achieves causal FPR 0.015 with event recall 0.507. Future temporal extrapolation is therefore not uniquely necessary for score conditioning; latent coordinate compression drives tail stability. (5) *Linear vs. non-parametric controls:* A linear projection forecaster (`PCA-LinPred-LN`) achieves $\gamma_2 = 0.95$, $D_{\text{KS}} = 0.233$, causal FPR 0.021, and event recall 0.452, whereas non-parametric isolation trees (`IsoForest`) collapse in event recall (0.149 at SPOT; 0.317 at f^*=0.01). Representation rank diagnostics confirm z_{ctx} and z_{tgt} occupy 5.6–6.1 and 8.5–14.0 of 32 dimensions, demonstrating that while LayerNorm geometry mechanically halves false alarms by bounding kurtosis, learned latent projection is strictly required to halve distribution drift and suppress false alarms to the 1.5% operational regime.

VII. Results and Discussion

A. Benchmark Performance on Continuous Telemetry

Table II reports the comparison across 33 continuous telemetry streams under the held-out calibration protocol of Section V-A (SPOT, $u = 0.98, q = 10^{-3}$), with buffered (a) and strictly causal (b) attribution reported separately. The principal finding is false-alarm control; the evidence for general detection superiority is inconclusive.

False Alarm Suppression on Held-Out Telemetry: Latent predictive scoring via Operator-Entropy JEPA and TS-JEPA substantially suppresses false alarms under static calibration: on 19 non-GHL streams, latent predictors achieve held-out nominal FPR 0.076–0.086 versus 0.300 for TimesNet and 0.307 for TranAD (four-fold reduction buffered, and 12–18$\times$ causal: 0.014–0.019 vs. 0.222–0.243). Across all 33 streams the paired buffered reduction is significant: OpEntropy JEPA achieves -0.144 vs. TimesNet (CI $[-0.241, -0.065]$, $p = 0.006$) and -0.142 vs. TranAD ($p = 0.006$); TS-JEPA yields -0.153 ($p = 0.002$) and -0.151 ($p = 0.003$). Dataset-clustering ($N = 7$) yields $p = 0.016$–0.031. The gain is stable under leave-one-dataset-out (LODO) resampling: across folds, macro buffered FPR stays 0.038–0.070 (OpEntropy) and 0.038–0.061 (TS-JEPA) vs. 0.084–0.176 (TimesNet) and 0.098–0.183 (TranAD). On MSL, TimesNet records lower buffered FPR than OpEntropy (0.133 vs. 0.191), while on Genesis, Metro, and SWAN reconstructors record zero FPR with zero F1, indicating inactivity rather than calibration.

Point-F1 and Continuous Ranking (PR-AUC) Uncertainty: False-alarm suppression does not establish superior anomaly detection. On the non-GHL subset TS-JEPA attains 0.234 Point-F1 / 0.323 PR-AUC versus 0.197/0.329 (TimesNet) and 0.130/0.287 (TranAD). Across all 33 streams paired differences are inconclusive: vs. TimesNet +0.012 F1 (CI $[-0.053, +0.075]$, p=0.54), -0.000 PR-AUC (p=0.39); vs. TranAD +0.051 F1 (p=0.88), +0.023 PR-AUC (p=0.16). Between-stream SDs match the means (SMAP 0.415 ± 0.389 for TS-JEPA vs. 0.289 ± 0.334 for TimesNet). Only SMAP exceeds constant-alarm reference $\text{F1}_{\text{all-pos}}$=0.236 (0.415 and 0.401); MSL (0.237, 0.303 vs. 0.381), Daphnet, and SWAN fall below it. PR-AUC leads over prevalence are the primary signal (SMAP 0.504/0.525 at π=0.150; MSL 0.434/0.422 at π=0.255, vs. 0.455/0.311 for TimesNet).

Regimes of Applicability: Spacecraft telemetry (SMAP, MSL) shows the largest PR-AUC lead over prevalence and reconstructors, though F1 margins sit within between-stream SD and labels carry known defects [32]. Quiet faults are filtered with ambient noise (Table I(d)). Process noise and continuous ramps (GHL, Metro) defeat all detectors (0.000–0.022 F1), and on high-prevalence streams (Daphnet π=14.8%, SWAN π=27.9%) all detectors score near or below constant alarm.

B. Evaluating Physical Inductive Priors: Hypothesis Testing and Null Results

Does adding physics-inspired regularizers (Koopman entropy, Reynolds stress, potential flow) improve detection over unconstrained `TS-JEPA`? The 33-stream data do not support an aggregate gain: paired differences against TS-JEPA are

TABLE I

ABLATION STUDIES. (A) MATCHED-CAPACITY REGULARIZER CONTROLS ON FIXED BACKBONES, 19 NON-GHL STREAMS (POINT-F1 = STREAM-MACRO MEAN ± SEED SD; HELD-OUT FPR ON UNSEEN NOMINAL DATA). (B) DYADIC-SHELL FRONT-END ON/OFF UNDER IDENTICAL TS-JEPA CONFIGURATION ACROSS FIVE MULTIVARIATE EXPLORATORY STREAMS (INCLUDING DIAGNOSTIC STREAMS SMD AND GECCO). (C) FACTORIAL AND STRENGTH SWEEP OF THE ENTROPY TERM (λ_{ENT}) AND ENERGY BARRIER (β_{NORM}) ON SIX STREAMS AT THREE SEEDS: OPERATOR RANK, OPERATOR CONDITION NUMBER, CONTEXT/TARGET REPRESENTATION RANK, LOSS COEFFICIENT OF VARIATION, AND HELD-OUT OUTCOMES (BUFFERED AND CAUSAL). (D) POINT RECALL UNDER INJECTED SINGLE-CHANNEL FAULTS ON SIX STREAMS: LATENT GATING REGIMES (UNIFORM, VARIANCE GATE, MAD GATE) AND OBSERVATION-SPACE REFERENCE MONITORS (`LOWVAR`, `DEV_MAXZ`); LAST ROW IS MEAN NOMINAL FPR. (E) CONTROLLED MECHANISM-DISCRIMINATION STUDY ACROSS 12 BACKBONE ARMS ON ALL 19 NON-GHL STREAMS (FIVE SEEDS; SPOT u=0.98, q=10^{-3}, NOMINAL CALIBRATION QUANTILE q^*=0.05, AND MATCHED REALIZED TEST FPR $f^* \in \{0.01, 0.05\}$).

(a) Regularizer Controls on Fixed Architectures

Model / Control	Point-F1 (buffered)	PR-AUC	Held-out FPR	Event Recall
TS-JEPA (Unconstrained)	0.234 ± 0.030	0.323	0.076	0.481
Operator-Entropy JEPA	0.237 ± 0.013	0.323	0.086	0.506
$\lambda_{\text{ent}} = 0$ control	0.244 ± 0.024	0.347	0.083	0.518
Reynolds-Stress JEPA	0.207 ± 0.052	0.307	0.078	0.434
$\alpha_{\text{stress}} = 0$ control	0.203 ± 0.059	0.311	0.066	0.414
Potential-Flow JEPA	0.089 ± 0.036	0.201	0.091	0.303
no-regime control ($K_r = 1$)	0.129 ± 0.061	0.265	0.081	0.392
TimesNet (Baseline)	0.197 ± 0.018	0.329	0.300	0.583
TranAD (Baseline)	0.130 ± 0.004	0.287	0.307	0.490

(b) Dyadic-Shell Front-End On/Off

Stream	Standard Conv	Dyadic Shells	Δ
SMAP `P-3`	**0.294**	0.261	−0.033
MSL `M-1`	0.002	**0.052**	+0.050
SMD `machine-1-2`	0.000	**0.177**	+0.177
Daphnet `S01R01E1`	0.144	**0.165**	+0.021
GECCO `water_quality`	0.000	0.001	+0.001
Macro Mean	0.088	**0.131**	+0.043

(c) Factorial and Strength Sweep of the Operator Regularizer

			Operator / representation / stability				Buffered			Causal		
Arm	λ_{ent}	β_{norm}	RankMe($\mathbf{K}$)	$\kappa(\mathbf{K})$	RankMe($z_{\text{ctx}}/z_{\text{tgt}}$)	$\text{CV}_{\mathcal{L}}$	FPR	F1	ER	FPR	F1	ER
Neither	0	0	31.83	1.235	5.15/10.04	1.013	0.066	0.084	0.393	0.022	0.028	0.516
Entropy only	0.5	0	31.97	1.088	5.25/10.07	0.977	0.092	0.083	0.429	0.023	0.023	0.456
Barrier only	0	1	31.83	1.234	5.18/10.02	1.010	0.084	0.085	0.440	0.025	0.026	0.472
Both	0.05	1	31.89	1.180	5.17/9.96	1.004	0.104	0.077	0.381	0.024	0.022	0.413
Both	0.10	1	31.91	1.156	5.34/9.97	1.001	0.093	0.094	0.444	0.023	0.023	0.437
Both (deployed)	0.50	1	31.97	1.083	5.21/9.94	0.976	0.081	0.076	0.373	0.023	0.024	0.452
Both	1.00	1	31.99	1.054	5.28/9.92	0.955	0.074	0.083	0.456	0.023	0.026	0.544
Both	2.00	1	32.00	1.034	5.28/9.95	0.917	0.082	0.088	0.373	0.022	0.025	0.433

(d) Quiet-Fault Point Recall

Injected fault	Unif.	Var.	MAD	`lowvar`	`dev_maxz`
Flatline	0.021	0.043	0.058	0.182	0.087
Flatline (mid-var.)	0.024	0.048	0.068	0.317	0.087
Stuck sensor	0.021	0.043	0.058	0.182	0.087
Dropout	0.020	0.043	0.058	0.182	0.087
Low-amp. step	0.030	0.049	0.062	0.182	0.087
$+5\sigma$ spike	0.023	0.042	0.059	0.182	0.108
Mean nom. FPR	0.019	0.069	0.055	0.149	0.141

(e) Controlled Mechanism-Discrimination Study (19 Non-GHL Streams)

	Tail/Drift		Held FPR		Point-F1		PR-AUC		Causal Event Recall			
Arm	γ_2	D_{KS}	Caus.	Buf.	Caus.	Buf.	Caus.	Buf.	SPOT	q^*=0.05	f^*=0.01	f^*=0.05
`TS-JEPA`	1.18	0.198	0.015	0.080	0.105	0.234	0.270	0.323	0.471	0.774	**0.439**	0.604
`OpEntropy`	1.11	0.201	0.021	0.083	0.112	**0.238**	0.252	0.322	0.512	0.771	0.404	0.565
`JEPA-CtxRecon`	2.49	0.199	0.015	0.108	0.118	0.208	0.246	0.277	0.507	**0.779**	0.409	0.572
`JEPA-NoEMA`	0.93	**0.194**	**0.012**	0.113	0.145	0.234	0.265	0.314	0.503	0.765	0.393	0.573
`TS-JEPA-NoGate`	0.90	0.202	0.014	0.080	0.123	0.235	0.268	0.324	0.469	0.768	0.415	0.582
`PCA-LinPred-LN`	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	**0.624**
`TCN-ObsPred-LN`	1.14	0.386	0.117	0.186	0.080	0.131	0.240	0.263	0.480	0.683	0.351	0.535
`IsoForest`	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
`TCN-ObsPred`	6.08	0.460	0.246	0.310	0.139	0.144	0.271	0.310	0.544	0.629	0.169	0.382
`TCN-ObsRecon`	6.28	0.445	0.245	0.303	0.133	0.135	0.262	0.301	0.528	0.620	0.163	0.380
`TimesNet`	5.54	0.438	0.243	0.299	**0.210**	0.194	**0.293**	**0.329**	**0.589**	0.618	0.217	0.407
`TranAD`	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

+0.002 Point-F1 for Operator-Entropy (CI $[-0.025, +0.029]$, p=0.71; PR-AUC +0.002, p=0.86; FPR +0.009, p=0.75), −0.015 for Reynolds-Stress ($[-0.042, +0.007]$, p=0.61), and −0.081 for Potential-Flow (p=0.033 unadjusted). Physical regularizers provide no aggregate detection advantage over unconstrained latent prediction; the entropy term preserves operator conditioning without altering representation rank or detection.

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Strictly causal results provide the operational evaluation; buffered results are retrospective. Table II(b) reports strictly causal streaming. On 19 non-GHL streams, TS-JEPA reduces held-out nominal FPR to 0.0136 versus 0.2427 (TimesNet) and 0.2219 (TranAD), an ≈18× suppression. On 14 GHL streams, causal FPR falls below 0.009, but causal Point-F1 is near-zero (≤ 0.006 vs. 0.022 TimesNet). Causal Point-F1 is 0.1118±0.0412 (TS-JEPA) versus 0.2091±0.0265 (TimesNet), and causal event recall is 46.55% vs. 58.82%: without lookahead buffers, latent scoring operates as an ultra-conservative filter trading sensitivity for false-alarm suppression. Dynamic SPOT on five incident streams suppresses false alarms up to 21× under drift.

D. Horizon Length, Score-Emission Latency, and False-Alarm Control

At S=64, scores emit 63 steps after window opening (10.5 min on 10-s telemetry; 63 h on hourly Metro data). Table III(a) retrains models at $S \in \{1, 4, 16, 32, 64\}$ on five streams (three seeds). Shortening horizon preserves false-alarm control: at S=1 (no emission lag), OpEntropy holds causal FPR at 0.005 (TimesNet 0.025) with event recall 0.729 (TimesNet 0.895). Causal FPR for reconstructors rises steeply with horizon (0.025 → 0.213), while latent models remain at 0.003–0.020. Buffered attribution inflates FPR with S (0.005 → 0.099 for OpEntropy). Where rapid response matters, S=1 to 4 is the natural operating regime.

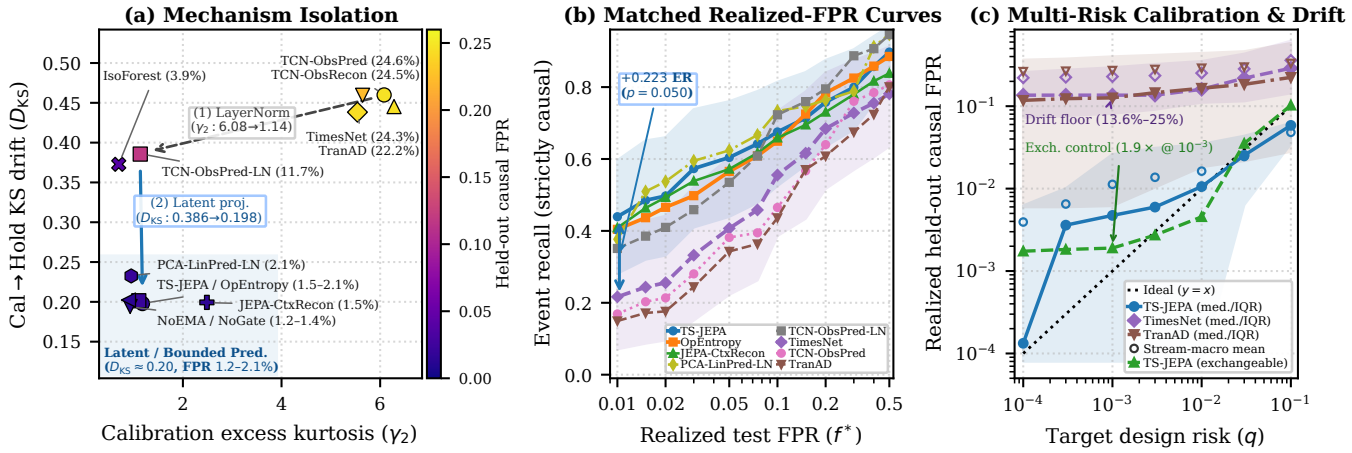


Fig. 2. Empirical isolation of residual conditioning, operating curves, and extreme-value calibration across 19 non-GHL telemetry streams (five seeds, strictly causal trailing inference). **(a) Mechanism isolation:** Calibration excess kurtosis (γ_2) versus calibration-to-holdout Kolmogorov–Smirnov drift (D_{KS}), colored by held-out causal FPR at SPOT $q{=}10^{-3}$. Adding a $D{=}32$ LayerNorm bottleneck to an observation forecaster (`TCN-ObsPred` → `TCN-ObsPred-LN`) mechanically compresses tail kurtosis ($\gamma_2 : 6.08 \rightarrow 1.14$) and halves causal FPR (24.6% → 11.7%), whereas learned latent projection (`TCN-ObsPred-LN` → `TS-JEPA`, `OpEntropy`, `JEPA-CtxRecon`) halves temporal distribution drift ($D_{\mathrm{KS}} : 0.386 \rightarrow 0.198$, $p{=}0.0001$) and suppresses causal FPR to 1.5%–2.1%. **(b) Matched realized test-FPR operating curves:** Strictly causal event recall across realized test false-alarm rates $f^* \in [0.01, 0.50]$ (shaded bands show 1,000-resample stream-bootstrap 95% CIs for `TS-JEPA` and `TimesNet`). Latent prediction holds a +0.223 event recall advantage at $f^*{=}0.01$ (0.439 vs. 0.217, $p{=}0.050$) that narrows for $f^* \geq 0.10$. **(c) Multi-risk calibration reliability diagram:** Realized held-out causal FPR (median solid/dashed lines, shaded IQR bands, and hollow stream-macro means) across target design risks $q \in [10^{-4}, 10^{-1}]$. Static thresholding overshoots the $y{=}x$ diagonal across all architectures due to inter-block drift; under an exchangeable point-shuffle control (green curve), `TS-JEPA` collapses to 1.9× nominal risk (0.0019 at $q{=}10^{-3}$).

E. Detection Quality at Matched Operating Points

We fit empirical-quantile thresholds on nominal calibration budgets $q \in [10^{-4}, 0.9]$ (Table III(b)). At every causal budget, latent predictors raise fewer alarms than reconstructors at comparable or higher precision: at $q^*{=}0.05$, TS-JEPA achieves event recall 0.774 at 179.8 false alarms per 1,000 nominal steps (precision 0.257) versus 0.618 at 262.3 alarms (0.221) for TimesNet. Retrospectively buffered at $q^*{=}0.001$, the recall ordering reverses (0.486 vs. 0.583), while precision remains higher (0.277 vs. 0.194) and alarm volume lower (207.9 vs. 259.2). Budgets match calibration quantiles rather than realized out-of-sample false-alarm rates (at $q^*{=}0.05$, realized test FPR is 0.180 vs. 0.262). Because anomalies span multi-hundred steps (mean 402 in Daphnet, 1,102 in SMAP), continuous alarming at 30% FPR catches over 95% of segments by random coincidence ($1 - (1 - \mathrm{FPR})^L$); TimesNet's 58% recall at 30% nominal FPR falls within chance expectations.

To test whether latent scoring detects more events at genuinely equal alarm rates, we evaluate direct matched realized test-FPR operating points across all 19 non-GHL streams at $f^* \in \{0.01, 0.05, 0.10, 0.20\}$ (Fig. 2(b), Table I(e)). At ultra-low realized false-alarm rates ($f^* = 0.01$ strictly causal), TS-JEPA achieves event recall 0.439 versus 0.217 for TimesNet (paired difference +0.223, 95% bootstrap CI $[+0.007, +0.428]$, Wilcoxon $p = 0.050$) and 0.149 for TranAD (paired difference +0.290, 95% CI $[+0.085, +0.503]$, $p = 0.035$). At moderate realized FPR ($f^* \geq 0.10$ causal and across smear mappings), event recall differences narrow or favor reconstructors. When common support is restricted to $f^* \in [0.10, 0.50]$ without extrapolation, the causal TS-JEPA–TimesNet difference is +0.061 (95% dataset-bootstrap CI $[-0.153, +0.329]$, sign-permutation $p = 0.625$, 20 paired cells across four streams). Latent prediction holds a decisive detection lead specifically under stringent ultra-low false-alarm constraints ($f^* \leq 0.01$), whereas under relaxed budgets reconstructors recover comparable event coverage.

F. Automated Threshold Calibration Realities on Non-Stationary Telemetry

Does automated extreme-value thresholding adhere to its design risk q out of sample? It does not. At design risk $q{=}10^{-3}$, observed held-out FPR is 5.3–6.2% buffered and 1.0–1.5% causal for the JEPAs, which is a calibration failure under distribution shift: the latent models overshoot the target by one to two orders of magnitude and are only less wrong than the reconstructors (≈ 22% buffered, 16–18% causal). Table IV and Fig. 2(c) compare calibrators on identical score streams across all 33 streams under strictly causal trailing and buffered smear inference. The calibrator evaluated is the *static* peaks-over-threshold step of SPOT, a single threshold fit once on the calibration block. Three realities emerge:

1) **Causal vs. buffered smear overshoot:** Under causal trailing, TS-JEPA reaches held-out FPR 0.0101 (median 0.0002, IQR $[0.0000, 0.0025]$); buffered inference smears each alarm over $S{=}64$ steps and reaches 0.0526, below the $S \cdot q = 6.4\%$ smear ceiling. TimesNet and TranAD exceed that ceiling even causally (0.1795, 0.1601), which indicates distributional breakdown.
2) **Multi-risk calibration breakdown:** Across a multi-target sweep $q \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ (Fig. 2(c)), static thresholding overshoots nominal design targets across all models. For TS-JEPA, median causal FPR is 0.0001 (IQR $[0.0000, 0.0062]$) at $q{=}10^{-4}$, 0.0047 ($[0.0000, 0.0260]$) at $q{=}10^{-3}$, 0.0106 ($[0.0000, 0.0347]$) at $q{=}10^{-2}$, and 0.0585 ($[0.0349, 0.0622]$) at $q{=}10^{-1}$. TimesNet overshoots far more severely, with median causal FPR pinned at 0.1360

TABLE II

HELD-OUT CALIBRATION BENCHMARK (SPOT, $q{=}10^{-3}$, HELD-OUT NOMINAL FPR ON DATA THE THRESHOLD NEVER SAW). (A) RETROSPECTIVE BUFFERED ATTRIBUTION ($S{=}64$) AND (B) STRICTLY CAUSAL TRAILING ATTRIBUTION (ZERO LOOKAHEAD), REPORTED SEPARATELY: NON-GHL = STREAM-MACRO OVER 19 STREAMS AT FIVE SEEDS; GHL = 14 STREAMS AT THREE TO FIVE SEEDS; DS-MACRO F1 = UNWEIGHTED MEAN OF THE 7 DATASET MEANS; Δt = MEAN DELAY OVER DETECTED EVENTS ONLY. BEST IN **BOLD**, SECOND UNDERLINED. (C) PER-DATASET BUFFERED RESULTS: n STREAMS, K CHANNELS, ANOMALY PREVALENCE π, CONSTANT-ALARM REFERENCE $\mathrm{F1}_{\text{ALL-POS}}{=}2\pi/(1{+}\pi)$; F1 IS MEAN $\pm$ BETWEEN-STREAM SD ($\pm$ SEED SD FOR SINGLE-STREAM DATASETS). LEAVE-ONE-DATASET-OUT RANGES ARE GIVEN IN THE TEXT. (D) PUBLISHED BASELINES IN THE 11-DATASET, 130-STREAM SINGLE-SEED PANEL UNDER THE SAME CALIBRATION PROTOCOL (POINT-F1, DATASET-MACRO / STREAM MEAN).

	(a) Buffered, non-GHL (19 streams)					(b) Causal, non-GHL			GHL buffered (14)		Summary
Model	FPR ↓	F1 ↑	PR-AUC ↑	ER ↑	Δt ↓	FPR ↓	F1 ↑	ER ↑	FPR ↓	F1 ↑	DS-Macro F1 ↑
Latent-space predictive architectures (JEPAs)											
TS-JEPA (unconstrained)	0.076	0.234	0.323	0.481	80.3	0.014	0.112	0.466	0.005	0.000	**0.148**
Operator-Entropy JEPA	0.086	**0.237**	0.323	0.506	192.3	0.019	0.109	0.469	0.012	0.001	0.130
Reynolds-Stress JEPA	0.078	0.207	0.307	0.434	132.8	0.011	0.088	0.456	**0.003**	0.002	0.122
Potential-Flow JEPA	0.091	0.089	0.201	0.303	146.9	**0.010**	0.023	0.343	0.011	0.006	0.065
Observation-space reconstructive baselines											
TimesNet	0.300	0.197	**0.329**	**0.583**	**48.5**	0.243	**0.209**	**0.588**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.141	0.533	0.046	0.020	0.069
Non-neural linear baseline											
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	**0.002**	0.000	0.038

	(c) Dataset descriptors				Buffered Point-F1 (mean ± SD)				Buffered held-out FPR			
Dataset	n	K	π	$\mathrm{F1}_{\text{all-pos}}$	TS-JEPA	OpEnt	TimesNet	TranAD	TS-JEPA	OpEnt	TimesNet	TranAD
Daphnet	6	9	.148	.256	0.125±0.076	0.136±0.143	0.229±0.232	0.144±0.119	0.066	0.023	0.313	0.298
GHL	14	16	.013	.026	0.000±0.000	0.001±0.003	0.022±0.030	0.020±0.025	0.004	0.012	0.061	0.046
Genesis	1	18	.003	.006	0.017±0.023	0.007±0.017	0.000±0.000	0.000±0.000	0.069	0.065	0.000	0.000
MSL	4	55	.255	.381	0.237±0.259	0.303±0.189	0.158±0.146	0.151±0.149	0.147	0.191	0.133	0.241
SMAP	6	25	.150	.236	0.415±0.389	0.401±0.343	0.289±0.334	0.167±0.341	0.062	0.108	0.549	0.513
Metro	1	5	.001	.003	0.000±0.000	0.000±0.000	0.001±0.001	0.000±0.000	0.017	0.020	0.000	0.000
SWAN	1	38	.279	.437	0.241±0.150	0.060±0.069	0.000±0.000	0.000±0.000	0.009	0.000	0.000	0.000

(d) 130-stream cross-check (Point-F1, dataset-macro / stream mean): NCAD 0.058/0.088; DCdetector 0.070/0.097; Anomaly Transformer 0.091/0.082; TimesNet 0.129/0.151; TranAD 0.156/0.152; TS-JEPA 0.160/0.186.

TABLE III

(A) TARGET-HORIZON AND SCORE-EMISSION SWEEP ON FIVE STREAMS AT THREE SEEDS (S = TARGET HORIZON; LAG $= S{-}1$ STEPS; HELD-OUT FPR, EVENT RECALL (ER), AND MEAN DELAY Δt IN STEPS UNDER STRICTLY CAUSAL TRAILING ATTRIBUTION, WITH BUFFERED OPENTROPY FPR/ER FOR CONTRAST). (B) DETECTION QUALITY AT MATCHED NOMINAL CALIBRATION BUDGETS q^* ON 19 NON-GHL STREAMS (STREAM-AVERAGED; ER = EVENT RECALL; P = POINT PRECISION; FA = FALSE ALARMS PER 1,000 NOMINAL STEPS). BUDGETS MATCH CALIBRATION QUANTILES, NOT REALIZED TEST FPR.

(a) Target-Horizon and Score-Emission Sweep

		OpEntropy, causal			TS-JEPA, causal		TimesNet, causal		OpEntropy, buf	
S	Lag	FPR	ER	Δt	FPR	ER	FPR	ER	FPR	ER
1	0	0.005	0.729	76.3	0.004	0.524	0.025	0.895	0.005	0.729
4	3	0.007	0.624	179.3	0.006	0.395	0.031	0.910	0.020	0.481
16	15	0.003	0.257	200.3	0.003	0.257	0.061	0.700	0.043	0.391
32	31	0.011	0.219	397.9	0.003	0.276	0.098	0.657	0.057	0.376
64	63	0.020	0.410	209.8	0.004	0.200	0.213	0.614	0.099	0.367

(b) Detection at Matched Nominal Budgets q^*

		TS-JEPA			OpEntropy			TimesNet			TranAD		
Regime	q^*	ER	P	FA	ER	P	FA	ER	P	FA	ER	P	FA
Causal	0.001	0.499	0.261	70.0	0.557	0.232	103.1	0.590	0.219	222.4	0.533	0.156	209.6
Causal	0.050	0.774	0.257	179.8	0.771	0.234	193.3	0.618	0.221	262.3	0.602	0.159	269.5
Buffered	0.001	0.486	0.277	207.9	0.523	0.264	247.9	0.583	0.194	259.2	0.516	0.130	262.6
Buffered	0.050	0.560	0.246	258.6	0.627	0.248	314.9	0.612	0.196	287.3	0.596	0.145	295.3

across $q \in \{10^{-4}, 10^{-3}\}$ (IQR $[0.0729, 0.2576]$ and $[0.0815, 0.2945]$) before reaching 0.1603 ($[0.1108, 0.3510]$) at $q{=}10^{-2}$ and 0.2889 ($[0.1440, 0.6186]$) at $q{=}10^{-1}$. TranAD similarly yields median causal FPR 0.1177, 0.1262, 0.1651, and 0.2236. Static calibration fails to maintain design risk across models because inter-block distribution drift shifts score quantiles between calibration and test blocks.

3) **Parity with other rules:** Empirical p99.5, validation-maximum, and split conformal thresholding ($\alpha{=}10^{-3}$) also overshoot because drift violates exchangeability, and the kurtosis-adaptive multiplier raises TS-JEPA buffered FPR from 0.0526 to 0.1768.

Empirical tail diagnostics on nine non-GHL streams confirm tail scarcity: median exceedances above $u = p_{98}$ is only 21 (16–70), and Grimshaw yields degenerate $\xi = -0.9$ in 33% of TS-JEPA and 52% of TimesNet fits. To separate drift from sample scarcity, an exchangeable permutation control samples calibration and evaluation splits from pooled nominal blocks: causal TS-JEPA FPR drops from 0.0121 ($12.1\times$ overshoot) to 0.0019 ($1.9\times$ overshoot), and TimesNet drops from 0.0872 to 0.0037 (96% reduction; block permutation $L{=}64$ yields $5.8\times$ overshoot for TS-JEPA). This indicates inter-block drift dominates calibration failure under exchangeability, with autocorrelation secondary. Across $N = 1080$ dumped cells, KS drift between calibration and test blocks predicts realized FPR (Spearman $\rho = +0.377$, 95% CI $[+0.314, +0.433]$). In the controlled-drift stress test (Table IV(b), 16 streams, design $q{=}10^{-3}$), static SPOT sits at 0.061 FPR without injection; rolling calibrators hold risk to 1%–7% under discrete shifts but fail under continuous ramps (0.33–0.39 FPR).

VIII. CONCLUSION AND FUTURE WORK

We formulated Operator-Entropy JEPA for multi-sensor anomaly detection and evaluated false-alarm control, discriminative detection, and operational calibration across 33 telemetry streams under a strict held-out protocol:

1) **False-Alarm Control and Matched Operating Points:** Latent predictive scoring suppresses nominal false alarms under static calibration four-fold buffered and 12–18×

TABLE IV

(A) CALIBRATOR COMPARISON ON IDENTICAL SCORE STREAMS (33 STREAMS, CELL-WEIGHTED MEANS WITH STREAM-MACRO IN PARENTHESES; UP TO FIVE SEEDS). DESIGN RISK q=10^{-3}=0.1%. OVERSHOOT $> q$ IS THE SHARE OF CELLS WHOSE 95% CI LOWER BOUND EXCEEDS $q = 10^{-3}$. BUFFERED SMEAR INFLATES FPR BY PAINTING ALARMS OVER $S = 64$ STEPS; CAUSAL TRAILING REFLECTS STRICT STEP-LEVEL ALARMS. (B) CONTROLLED-DRIFT STRESS TEST: ACHIEVED HELD-OUT FPR (DESIGN $q = 10^{-3}$) OF REAL HELD-OUT RESIDUAL STREAMS UNDER INJECTED SHIFTS, AVERAGED OVER 16 STREAMS WITH ≥ 200 CALIBRATION AND EVALUATION WINDOWS. ROLLING VARIANTS RECALIBRATE PER EVALUATION BLOCK.

Model Architecture	Calibrator Rule	Causal FPR	Overshoot $> q$	Buffered FPR	Point-F1 (buf)
TS-JEPA (Control)	Standard SPOT ($q = 10^{-3}$)	0.0101 (0.0092)	7%	0.0526 (0.0457)	0.1577
	SPOT + Kurtosis ($\beta = 0.167$)	0.0108 (0.0094)	7%	0.1768 (0.1515)	0.1694
	Empirical 99.5th Percentile	0.0169 (0.0157)	21%	0.0682 (0.0595)	0.1598
	Validation Maximum	0.0090 (0.0080)	6%	0.0553 (0.0478)	0.1591
Operator-Entropy	Standard SPOT ($q = 10^{-3}$)	0.0152 (0.0144)	11%	0.0617 (0.0546)	0.1599
	SPOT + Kurtosis ($\beta = 0.167$)	0.0136 (0.0125)	9%	0.1239 (0.1064)	0.1441
	Empirical 99.5th Percentile	0.0238 (0.0229)	25%	0.0758 (0.0677)	0.1658
	Validation Maximum	0.0148 (0.0136)	7%	0.0656 (0.0576)	0.1627
TimesNet	Standard SPOT ($q = 10^{-3}$)	0.1795 (0.1611)	49%	0.2217 (0.1985)	0.1395
	SPOT + Kurtosis ($\beta = 0.167$)	0.1843 (0.1623)	46%	0.2281 (0.2011)	0.1352
	Empirical 99.5th Percentile	0.1914 (0.1752)	51%	0.2338 (0.2125)	0.1413
	Validation Maximum	0.1791 (0.1599)	48%	0.2263 (0.2032)	0.1410
TranAD	Standard SPOT ($q = 10^{-3}$)	0.1601 (0.1423)	46%	0.2217 (0.1963)	0.0936
	SPOT + Kurtosis ($\beta = 0.167$)	0.1722 (0.1513)	43%	0.2453 (0.2151)	0.0940
	Empirical 99.5th Percentile	0.1712 (0.1544)	51%	0.2343 (0.2093)	0.0962
	Validation Maximum	0.1580 (0.1392)	44%	0.2270 (0.2009)	0.0952

(b) Injected condition	Static SPOT	Static empirical	Rolling empirical	Rolling SPOT
Stationary (no injection)	0.061	0.162	0.016	0.047
Mean shift $+1.0\sigma$	0.087	0.216	0.011	0.043
Regime change $+1\sigma$	0.073	0.187	0.015	0.070
Variance scale 2.0	0.082	0.195	0.016	0.047
Autocorrelation 0.9	0.113	0.232	0.024	0.055
Continuous mean ramp	0.075	0.186	0.326	0.385

causally against TimesNet and TranAD across leave-one-dataset-out folds. At ultra-low matched realized FPR ($f^* = 0.01$ causal), TS-JEPA reaches event recall 0.439 versus 0.217 for TimesNet and 0.149 for TranAD ($p \leq 0.050$), while at moderate realized FPR ($f^* \geq 0.10$) event recall differences narrow or favor reconstructors. PR-AUC and Point-F1 remain statistically comparable.

2) **Mechanism Isolation and Regularizer Equivalence:** Contrasting 12 controlled backbones decouples score compression from representation learning: LayerNorm bottlenecks mechanically bound kurtosis and halve FPR, but latent projection is strictly required to halve calibration-to-holdout distribution drift ($0.386 \rightarrow 0.198$) and suppress false alarms an additional eight-fold ($0.117 \rightarrow 0.015$). Adding physical regularizers yields formal TOST equivalence within ± 0.05 Point-F1 ($p \leq 0.0054$), functioning as structural constraints rather than empirical performance drivers.
3) **Calibration Realities Under Non-Stationarity:** Static extreme-value thresholding consistently overshoots design targets across multi-target sweeps ($q \in [10^{-4}, 10^{-1}]$) because inter-block distribution drift violates exchangeability, accounting for 84–96% of causal overshoot under exchangeable controls. Rolling thresholding tracks discrete shifts but fails under continuous drift ramps.
4) **Latency Limits and Fault Boundaries:** Predictive scoring incurs an $S-1$ emission lag (10.5 min at S=64 on 10-s telemetry). While zero-lag monitoring (S=1) preserves false-alarm control (0.005 vs. 0.025), quiet flatlines remain undetected without high-alarm auxiliary variance monitors.

Open items include adaptive thresholding tracking within-block drift and low-latency architectures for quiet fault modes.

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